Resultant multiplicities at the fifteen prime Fricke levels

A source audit, geometric proof, and exact computation

Research memorandum prepared for David Jordan

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Abstract

The corrected general identity holds in every one of the 1,127 tested pairs, comprising all fourteen odd levels in Nakaya's set and level 2, for every prime $5 \le p \le 400$, $p \neq N$. A geometric proof for every permitted prime is given below, using norms and divisors on the smooth modular curve; it does not assume that its plane image is smooth. Two qualifications are essential: unramified elliptic factors must be removed before defining A_N, B_N, and level 2 needs two different CM exponents. Morton's four original cases are already proved; priority for the extension to the other eleven levels is not established by this literature search.

1 Source transcription and the original four cases

The following is a transcription of Conjecture 4 in the supplied Nakaya arXiv v1, Section 4.1, p. 18. All equalities are modulo p, for primes $p \ge 5$, $p \neq N$. Write $s_N = ss_p^{(N*)}(Y)$, $\delta = (1 - (\frac{-3}{p}))/2$, $\epsilon = (1 - (\frac{-1}{p}))/2$, $\nu = (1 - (\frac{-2}{p}))/2$, and $\mu_N = (1 - (\frac{-N}{p}))/2$.

$$\begin{aligned}
Y^{\epsilon}(Y-256)^{\nu}V_p^{(2*)}(Y) &= (Y+144)^{2\delta}(Y-648)^{\epsilon}s_2^2,\\
Y^{\delta}(Y-108)^{\delta}V_p^{(3*)}(Y) &= (Y+192)^{2\delta}(Y^2-576Y-1728)^{\epsilon}s_3^2,\\
(Y^2-44Y-16)^{\mu_5}V_p^{(5*)}(Y) &= (Y^2+216Y+144)^{2\delta}(Y^2-540Y-6480)^{\epsilon}s_5^2,\\
(Y+1)^{\mu_7}(Y-27)^{\mu_7}V_p^{(7*)}(Y) &= (Y^2+224Y+448)^{2\delta}\\
&\quad\cdot(Y^4-528Y^3-9024Y^2-5120Y-1728)^{\epsilon}s_7^2.
\end{aligned}$$

The page was checked both by text extraction and visual rendering. The code verifies the four displayed correction polynomials explicitly, and verifies the identities at all 302 permitted pairs in these four levels up to 400. There is no discrepancy with Nakaya's formulas.

1.1 Morton's exact scope

Level	Result in supplied Morton v1	Verified page	Hypothesis
5	Theorem 1, equations (6)–(7)	3	prime $p \ge 7$
7	Theorem 3, equations (13)–(14)	12	prime $p \ge 5$, $p \neq 7$
2, 3	Theorem 5, equations (26)–(27)	21	prime $p \ge 5$

These are exactly all four levels in the original Conjecture 4. For level 5, $p \ge 7$ is the same permitted domain as $p \ge 5$, $p \neq 5$. Morton's exceptional primes in the proofs are checked separately; they are not additional exclusions from the theorems. The introduction, p. 2, explicitly identifies the three theorems as covering all four cases.

Pagination limitation. The uploaded file is arXiv:2105.14532v1, with 27 pages, not the 26-page journal typesetting. The journal citation is *International Journal of Number Theory, 19*(4), 803–828 (2023), DOI 10.1142/S1793042123500409. The theorem numbers and pages in the table were verified in the uploaded preprint. The exact journal page for each theorem, and unchanged numbering in the final typesetting, are **not verified**; the accessible publisher preview contains only the opening page. No journal page was inferred by adding a fixed offset.

2 A uniform statement and its proof

Let $\mathfrak{S} = \{2,3,5,7,11,13,17,19,23,29,31,41,47,59,71\}$ and use Nakaya's constant terms $104,42,16,9,0,\dots,0$. Write

$$R_N(X,Y) = X^2 - a_N(Y)X + b_N(Y), \quad D_N = a_N^2 - 4b_N, \quad Q_{N,c} = R_N(c,Y).$$

All gcds and factorizations in the following definitions are taken in $\mathbb{Q}[Y]$, with monic factors. For $c = 0, 1728$, put

$$U_{N,c} = \prod_{\substack{f \text{ monic irreducible},\ f \mid Q_{N,c} \\ f \mid a_N - 2c}} f^{v_f(Q_{N,c})}. \tag{1}$$

Thus $U_{N,c}$ is the full part of $Q_{N,c}$ supported on its gcd with $a_N - 2c$, not just that gcd. Define the monic polynomials A_N, B_N, E_N, H_N by

$$Q_{N,0} = U_{N,0}A_N^3, \qquad Q_{N,1728} = U_{N,1728}B_N^2, \qquad D_N = E_N H_N^2, \tag{2}$$

where E_N is squarefree. Equivalently, E_N is the product of the irreducible factors of D_N having odd exponent. These polynomials belong to $\mathbb{Z}[Y]$: they are monic rational factors of monic integer polynomials, so Gauss's lemma applies.

For an odd prime $p \geq 5$, let

$$\delta = \frac{1 - \left(\frac{-3}{p}\right)}{2}, \quad \epsilon = \frac{1 - \left(\frac{-1}{p}\right)}{2}, \quad \mu_N = \frac{1 - \left(\frac{-N}{p}\right)}{2}, \quad \nu = \mu_2.$$

When $p \neq N$, set

$$C_{N,p}(Y) = \begin{cases} Y^{\epsilon}(Y-256)^{\nu}, & N = 2, \\ E_N(Y)^{\mu_N}, & N \text{ odd}. \end{cases} \tag{3}$$

Bars below mean coefficientwise reduction modulo p.

Theorem 1 (Resultant multiplicities for the prime Fricke levels)**.** *Let $N \in \mathfrak{S}$, and let $p \geq 5$ be prime with $p \neq N$. Put*

$$V_{N,p}(Y) = \mathrm{Res}_X(ss_p(X), \overline{R_N}(X,Y)), \qquad S_{N,p} = \mathrm{rad}(V_{N,p}),$$

where the radical is monic. Then $S_{N,p} = ss_p^{(N*)}$ and

$$\boxed{\overline{C_{N,p}}\, V_{N,p} = \overline{A_N}^{\,2\delta}\, \overline{B_N}^{\,\epsilon}\, S_{N,p}^{;2} \quad \text{in } \mathbb{F}_p[Y].} \tag{4}$$

For every monic irreducible factor f of $V_{N,p}$, its exact exponent is

$$v_f(V_{N,p}) = 2 + 2\delta\, v_f(\overline{A_N}) + \epsilon\, v_f(\overline{B_N}) - v_f(\overline{C_{N,p}}). \tag{5}$$

In particular, reduction multiplicities in the correction factors must be retained. If $d_p = \deg ss_p$, then $\deg V_{N,p} = (N+1)d_p$ and its leading coefficient is 1. There is no unspecified scalar in (4).

2.1 The norm lemma, including singular plane points

Lemma 1. *Let k be algebraically closed of characteristic different from 2, let C be a smooth curve with a nonidentity involution w, and let $\pi : C \to D = C/\langle w\rangle$. Suppose Y is a local coordinate on D, j is regular above the point under consideration, and*

$$R(X,Y) = (X - j)(X - j \circ w) = \mathrm{Norm}_{k(C)/k(D)}(X - j).$$

For any finite set $J \subset k$, put $s(X) = \prod_{\alpha\in J}(X - \alpha)$. Then

$$\mathrm{ord}_{Y=y} \mathrm{Res}_X(s, R) = \sum_{P\in\pi^{-1}(y)} \mathrm{ord}_P(s(j)) = \sum_{\substack{P\in\pi^{-1}(y)\\ j(P)\in J}} e_j(P). \tag{6}$$

The sums run over distinct points of the normalization, including a fixed point only once.

Proof. The product formula for the resultant gives $\mathrm{Res}_X(s, R) = \prod_{\alpha\in J} R(\alpha, Y) = \mathrm{Norm}(s(j))$; the possible signs cancel because the norm has degree 2. At an unramified point of D the completed algebra is a product of two power-series rings, and the order of the norm is the sum of the two orders. At a branch point, tame linearization gives parameters t, u with $w(t) = -t$ and $u = t^2$. If $s(j) = t^m v(t)$ with v a unit, its norm is $(-1)^m u^m v(t)v(-t)$, which has order m in u. This proves the first equality. Squarefreeness of s gives the second, since the order of $j - j(P)$ is $e_j(P)$. No regularity of the plane curve $R = 0$ has been used. □

2.2 Integral models and the three correction divisors

Proof of Theorem 1. Work over $R = \mathbb{Z}_{(p)}$. Use the smooth coarse modular curve $\mathcal{C} = X_0(N)_R$, its Fricke involution w, and its quotient $\mathcal{D} = \mathcal{C}/\langle w\rangle$. Since $p \nmid 6N$, the moduli stabilizers and the quotient involution are tame. Jordan's Section 2 and Proposition 5.1 identify $\mathcal{D}$ with $\mathbb{P}^1_R$ by $Y = j^*_N$, with its cusp at infinity. The two cusps of $\mathcal{C}$ are distinct and exchanged by w.

On the complement of those cusps, j is regular and $\pi : \mathcal{C} \to \mathcal{D}$ is finite locally free of degree two. Its characteristic polynomial is

$$\mathrm{Norm}(X - j) = X^2 - \mathrm{Tr}(j)X + \mathrm{Norm}(j).$$

Nakaya's Proposition 4 identifies this polynomial with R_N on the generic fibre. Both sides are integral sections of a torsion-free algebra, so the equality holds over R and after reduction. This is the required specialization statement; it is stronger and safer than assuming that the plane model is smooth.

For completeness, the plane model does have the correct function field in every permitted characteristic. The pole orders of j at the two cusps are 1 and N, whereas those of $j \circ w$ are N and 1. Their leading coefficients are units, so $j \neq j \circ w$ also after reduction. Hence $k(\mathcal{C}_k) = k(Y, j)$ and the smooth modular curve is the normalization of $R_N = 0$. The map j is separable: away from the elliptic points the level-forgetting morphism is etale. Its degree need not be prime to p for this conclusion.

Let T_3 be the reduced divisor of points above $j = 0$ with $e_j = 3$, and let T_2 be the analogous reduced divisor above $j = 1728$ with $e_j = 2$. At every other finite point $e_j = 1$. These descriptions hold on the generic and special fibres. Indeed, over an elliptic curve the choices of a cyclic subgroup of order N form a finite etale scheme; the coarse map is the tame quotient by $\mathrm{Aut}(E)/\{\pm 1\}$. A nonfixed subgroup has orbit size 3 or 2, respectively; a fixed subgroup has orbit size 1. This also proves that the ramified and unramified parts of each elliptic fibre specialize with their multiplicities unchanged on $\mathcal{C}$.

Over characteristic zero, an isogeny with source and target both of invariant 0, or both of invariant 1728, is an endomorphism after identifying the curves. The endomorphism ring is commutative, so its kernel is stable under all automorphisms. Conversely, a subgroup stable under all automorphisms gives a quotient with the same exceptional invariant. Thus the support of $U_{N,c}$ in (1) is precisely the image of the unramified part of $j^{-1}(c)$. Applying Lemma 1 to X and $X - 1728$ gives

$$\operatorname{div}_0(A_N) = \pi_* T_3, \qquad \operatorname{div}_0(B_N) = \pi_* T_2 \tag{7}$$

on the generic fibre. These equalities persist on the special fibre. One can see this without dividing polynomials modulo p: after an etale base change split the reduced elliptic fibres into sections, and take the products of $Y - Y(P)$ over the appropriate sections. Their monic products are the reductions of the characteristic-zero polynomials in (7). Distinct sections of $\mathcal{C}$ may have the same image in $\mathcal{D}$; such repetitions are retained in these products.

Let F be the fixed-point divisor of w. Its image on $\mathcal{D}$ is the branch divisor. In characteristic zero the quadratic extension can be written

$$k(\mathcal{C}) = k(Y)(\sqrt{D_N(Y)}) = k(Y)(\sqrt{E_N(Y)}).$$

Therefore $\operatorname{div}_0(E_N) = \pi_* F$. There is no branch point at the cusp. Since the involution is tame and nonidentity on every fibre, its fixed scheme is finite etale over R: locally at a fixed point the action on a parameter is $t \mapsto -t$. In particular, the branch points remain distinct. This proves that $\overline{E_N}$ is squarefree for every $p \geq 5$, $p \neq N$, and that it still cuts out precisely the branch divisor. The factors in H_N^2 are not extra branch points.

For odd $N \geq 5$, the characteristic-zero Fricke-fixed points have CM by orders of discriminant $-4N$, and also $-N$ when $N \equiv 3 \pmod 4$. Their common CM field is $\mathbb{Q}(\sqrt{-N})$. Every special-fibre fixed point lifts, by the etaleness just proved. The CM reduction criterion consequently says that all these fixed points are supersingular when $\left(\frac{-N}{p}\right) = -1$, and none are supersingular when that symbol is 1. Equivalently, one can combine the unchanged total fixed-point count with the supersingular fixed-point count of Beneish and Wen, Proposition 3.16, with $D = N, m = 1$; both counts agree in the inert case and the latter is zero in the split case. No ramified case occurs because $p \neq N$.

At $N = 3$, the two branch values are $0, 108$, with j-values $0, 54000$, of discriminants $-3, -12$; the same conclusion holds with $\mu_3 = \delta$. At $N = 2$, the values $0, 256$ correspond to discriminants $-4, -8$, respectively, and have different supersingularity tests ϵ and ν. Thus in all cases the divisor of $\overline{C_{N,p}}$ is exactly the image of the supersingular part of F.

Now let Z be the reduced supersingular divisor on $\mathcal{C}_k$, $k = \overline{\mathbb{F}}_p$, and let S denote its reduced image on $\mathcal{D}_k$. Supersingularity is preserved by the degree-N isogeny, so Z is w-stable. The orbit count, with a fixed point counted once, is the divisor equality

$$\pi_* Z = 2S - \pi_* F_{\mathrm{ss}}. \tag{8}$$

On the other hand, the finite zero divisor of $ss_p(j)$ is

$$\operatorname{div}_0(ss_p(j)) = Z + 2\delta T_3 + \epsilon T_2.$$

Push this equality to $\mathcal{D}_k$ and apply the norm lemma, (7), and (8). It gives

$$\operatorname{div}_0(V_{N,p}) = 2S + 2\delta \operatorname{div}_0(\overline{A}_N) + \epsilon \operatorname{div}_0(\overline{B}_N) - \operatorname{div}_0(\overline{C}_{N,p}).$$

The support is precisely the set in Nakaya's definition, so S is the zero divisor of $S_{N,p} = ss_p^{(N*)}$. This proves the claimed equality of divisors, even when correction factors collide. Every polynomial involved is monic. Moreover $R_N(\alpha, Y)$ is monic of degree $N + 1$ for each α, since a_N and b_N are monic of degrees N and $N + 1$. Their product over the d_p supersingular invariants proves the degree, leading coefficient, and scalar-one assertions. Taking the coefficient of an irreducible divisor proves (5). □

2.3 What is general, and what is special to these levels

The norm and divisor argument works for every prime $N \neq p$, with the Fricke quotient in place of the Y-line. A single polynomial statement in a Hauptmodul requires genus zero, hence the restriction to $\mathfrak{S}$ here. This proof uses Jordan's good-reduction coordinate result and tame moduli description, but does not need his nodal genus comparison in Lemma 3.1. That comparison supplies an independent aggregate check, not a local multiplicity calculation.

For odd prime N, the numbers of unramified points above the two elliptic invariants are

$$u_3 = 1 + \left(\frac{-3}{N}\right), \qquad u_2 = 1 + \left(\frac{-1}{N}\right).$$

At $N = 2$ they are $u_3 = 0, u_2 = 1$. A formula valid at all prime levels uses the Kronecker symbol $u_2 = 1 + \left(\frac{-4}{N}\right)$ instead. Consequently

$$\deg A_N = \frac{N + 1 - u_3}{3}, \qquad \deg B_N = \frac{N + 1 - u_2}{2}, \qquad \deg E_N = 2g(X_0(N)) + 2.$$

For odd $N \geq 5$, Jordan's fixed-point formula (4.1) gives

$$\deg E_N = h_{\mathrm{ord}}(-4N) + [N \equiv 3 \pmod 4] h_{\mathrm{ord}}(-N) = 2d_N.$$

The same last equality holds for $N = 3$ with $d_3 = 1$. Taking degrees in (4) therefore recovers Jordan's Corollary 1.3. At $N = 2$ it gives

$$\deg S_{2,p} = \frac{p-1}{8} + \frac{3}{8}\left(1 - \left(\frac{-1}{p}\right)\right) + \frac{1}{4}\left(1 - \left(\frac{-2}{p}\right)\right).$$

This is a consequence of the stronger multiplicity formula, not a new claim of priority for the known degree formulas.

3 Checks on the formulation

3.1 Why the two corrections to a naive statement matter

The polynomial $\mathrm{rad}(D_N)$ usually contains much more than the branch divisor. For example,

$$D_2 = Y(Y - 256)(Y - 81)^2.$$

The factor $Y - 81$ is excluded from E_2. At this value two distinct points on the normalization can have the same j-value. They still contribute separately to the norm. Counting a repeated X-root as one point would be wrong.

Also, taking the whole square root of $R_N(1728, Y)$ gives the wrong B_N. At level 5,

$$R_5(1728, Y) = Y^2(Y^2 - 540Y - 6480)^2,$$

so $U_{5,1728} = Y^2$ and $B_5 = Y^2 - 540Y - 6480$. The omitted square is the image of two unramified elliptic points. The rational gcd definition in (1) distinguishes it from the ramified square factors.

Finally, the prescription $E_N^{\mu_N}$ cannot be used unchanged at $N = 2$. Its branch points have CM fields $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-2})$. For instance at $(N, p) = (2, 5)$, $\epsilon = 0$, $\nu = 1$; multiplying by the additional factor Y would give a false identity. The user's proposed common CM-field description is correct at odd levels, but requires this level-two exception. The factors Y at level 2 and at level 3 lie over unramified exceptional elliptic points, so no extra ramification factor should be inserted for them.

3.2 Geometric multiplicity rule

For a supersingular quotient value y, let $b(y)$ be 1 if its fibre is Fricke-fixed and 0 otherwise. Let $n_3(y)$ and $n_2(y)$ count the distinct points in that fibre where $e_j = 3$ and $e_j = 2$. The norm proof gives

$$m(y) = 2 - b(y) + 2n_3(y) + n_2(y).$$

Only supersingular elliptic fibres contribute: $n_3 = 0$ unless $\delta = 1$ and $n_2 = 0$ unless $\epsilon = 1$. A nonfixed fibre has two points, hence can have multiplicity $2, 3, 4, 5$, or 6; a fixed fibre has multiplicity $1, 2$, or 3. In particular, a factor of multiplicity 6 is legitimate, not a failure of the square formula. Formula (5) is its version over $\mathbb{F}_p$ for arbitrary irreducible factors.

3.3 Recovery of the four formulas

N	E_N	A_N	B_N
2	$Y(Y-256)$	$Y+144$	$Y-648$
3	$Y(Y-108)$	$Y+192$	$Y^2-576Y-1728$
5	$Y^2-44Y-16$	$Y^2+216Y+144$	$Y^2-540Y-6480$
7	$(Y+1)(Y-27)$	$Y^2+224Y+448$	see below

Here $B_7 = Y^4 - 528Y^3 - 9024Y^2 - 5120Y - 1728$. Substituting these into (4), with $\mu_3 = \delta$ and the separate level-two definition (3), yields precisely the four identities transcribed above, including the leading scalar 1.

4 Exact computation

The final, clean run used PARI/GP 2.15.4 (amd64, GMP 6.3.0) and completed in **11.624 seconds** of GP wall-clock time on the supplied environment. This excludes installation, literature retrieval, and report generation. No floating-point arithmetic enters any mathematical check. The random seed only fixes finite-field factorization choices.

N	Pairs	Failures	Flagged	$\deg A_N$	$\deg B_N$	$\deg E_N$
2	76	0	3	1	1	2
3	76	0	4	1	2	2
5	75	0	5	2	2	2
7	75	0	8	2	4	2
11	75	0	13	4	6	4
13	75	0	10	4	6	2
17	75	0	14	6	8	4
19	75	0	14	6	10	4
23	75	0	15	8	12	6
29	75	0	18	10	14	6
31	75	0	19	10	16	6
41	75	0	25	14	20	8
47	75	0	27	16	24	10
59	75	0	35	20	30	12
71	75	0	36	24	36	14
Total	1127	0	246			

The fourteen odd levels account for 1,051 pairs; level 2 contributes 76.

4.1 Reconstruction and independence of the checks

The code constructs the Hauptmodul as a quotient of a form in the correct Fricke eigenspace by $(\eta(\tau)\eta(N\tau))^k$. It asserts the eigenspace dimension $m+1$, the principal part q^{-1}, Nakaya's constant term, and integrality of the retained coefficients. It constructs a_N, b_N by eliminating their pole parts; no stored relation is an input. The positive-coefficient checks implement the two identities of Nakaya's Proposition 4, not a comparison with an external coefficient table (that proposition supplies no such table for all fifteen levels).

The remainder orders range from $O(q^{97})$, $O(q^{96})$ at level 2 to $O(q^{443})$, $O(q^{442})$ at level 71 for the sum and product, respectively. Thus at level 71 all positive coefficients through q^{442} and q^{441} are checked. Principal-part matching certifies the identity once modularity and the pole divisor are known; the extra coefficients are implementation safeguards.

The ordinary supersingular polynomial is computed by a binomial formula and cross-checked, for each of the 76 tested primes, against the radical of the Legendre Hasse-polynomial elimination. Each $V_{N,p}$ is computed by a resultant and fully factored. Its radical is obtained by a derivative/gcd routine with recursive pth-root extraction, independently of the predicted correction identity; that radical is also checked against the product of the distinct irreducible factors. This is not an independent moduli construction of $S_{N,p}$: it is precisely Nakaya's definition, as requested.

Every irreducible factor's actual exponent is compared to (5); the full monic polynomial identity is also checked, which detects any missing support or scalar error. Radical degrees are compared with Jordan's Corollary 1.3 (his level-two formula for $N=2$), and each radical is checked to divide $Y^{p^2}-Y$. In the supplied paper, Corollary 6.3 concerns nonrational quadratic factors, rather than the total degree. This audit uses Corollary 1.3 for the degree, as specified in Task B.3.

4.2 Collision primes: no exclusions

The phrase "a prime dividing $\mathrm{disc}_X R_N$" needs interpretation: this is a polynomial, monic here, so no prime divides all its coefficients. Also $\mathrm{disc}_Y(D_N)=0$ identically because D_N has square factors. The audit therefore uses the nonzero integer certificates

$$\mathrm{disc}(A_N),\ \mathrm{disc}(B_N),\ \mathrm{disc}(E_N), \mathrm{Res}(A_N,B_N),\ \mathrm{Res}(A_N,E_N),\ \mathrm{Res}(B_N,E_N),$$

plus $\mathrm{disc}(\mathrm{rad}_{\mathbb{Q}}\, D_N)$. Divisibility of the last integer detects collisions among distinct characteristic-zero discriminant roots, including singular-plane points. These tests concern possible collisions whether or not the affected roots are supersingular.

There are 246 flagged pairs: 239 have a collision in the A, B, E correction data and 7 have only a collision in the larger discriminant-root set. The latter are

$$(N,p) = (13,17), (17,13), (19,17), (23,13), (29,13), (41,37), (71,37).$$

Every one of the 246 identities passes. No permitted prime divides $\mathrm{disc}(E_N)$; exact factorizations of these integers have prime support contained in $\{2, N\}$ for these fifteen levels. Thus two true branch points never coalesce in the conjectured range. A branch point can meet an elliptic correction root, and two ramified points can have the same quotient coordinate.

Two explicit checks illustrate the need to retain reduction multiplicities:

- $(N,p) = (2,11)$: $V = Y(Y+1)^5$, $S = Y(Y+1)$, $A = B = Y+1$, $C = Y$. Both sides of (4) equal $Y^2(Y+1)^5$.
- $(N,p) = (11,5)$: $A = Y^2(Y-1)^2$ modulo 5, $\delta = 1$, $\epsilon = 0$, $\mu_{11} = 0$, and $V = Y^6(Y-1)^6$, $S = Y(Y-1)$. Each of the two factors has predicted and actual exponent $2 + 2 \cdot 2 = 6$.

All flagged prime lists appear in Appendix A; their gcd certificates and the individual factor matrices are in the accompanying files. No smoothness inference is made from the absence of a flag. The proof already accommodates singular plane points and collisions through normalization and divisor specialization.

4.3 Runs outside the conjecture

The package includes all 30 pairs with $p = 2, 3$ and all 13 additional diagonal pairs with $p = N \geq 5$, for 43 separate exploratory pairs. For $p = 2, 3$ the ordinary supersingular polynomial is taken to be X. Only the resulting polynomials and radicals are reported: the tame exponents are not asserted there.

For the diagonal odd levels, substituting $\mu_N = 1/2$ does not define the original polynomial identity. As a deliberately formal diagnostic, the code checks its squared version

$$\overline{E}_N V_{N,N}^2 \stackrel{?}{=} \overline{A}_N^{4\delta} \overline{B}_N^{2\epsilon} S_{N,N}^4.$$

It fails at all 13 diagonal levels. For example, at $(N, p) = (5, 5)$, $V = (Y + 3)^6$, $S = Y + 3$, $\overline{E}_5 = (Y + 3)^2$ and $\overline{A}_5 = (Y + 3)^2$; the two sides have exponents 14 and 12. These are not counterexamples to Theorem 1.

4.4 How to reproduce

From the extracted `nakaya_c4` directory, create `results` if necessary and run with **PARI/GP 2.15.4**:

```
mkdir -p results
gp -fq -s 1000000000 code/verify_c4.gp > results/run.log 2>&1
python3 code/audit_outputs.py
```

The driver refuses other GP versions, rather than silently bypassing the reported Hauptmodul dimension issue. This fulfils the pinned-version option; a 2.17.2 fix is not claimed. To change the survey bound, use a small driver file containing

```
P_MAX=1000;
read("code/verify_c4.gp");
```

For function-only loading set `C4_NO_RUN=1` before `read`. The functions `nk_haupt`, `nk_faber`, `nk_relation`, `nk_ss`, and `nk_rad` adapt Jordan's supplied verification code. The new multiplicity audit, extraction of correction factors, collision certificates, and outside-domain diagnostics are labelled separately in the source. `verify_duality.gp` was not present in the uploaded archive.

5 Prior work and novelty

This is a bounded literature search, not a proof that no earlier general formula exists. Searches covered the two named papers, visible citing-paper lists, targeted queries for the resultant identity, and the requested Aricheta, Sakai, Tsutsumi, and Kaneko–Zagier literature. Technical claims below rely on the original papers or publisher abstracts, not automated literature summaries.

Source	What was established or inspected	Overlap	Verification
Nakaya (2019)	Proposition 4 constructs the two monic coefficient polynomials. Section 4.1, Conjecture 4 states the four identities; Remark 2 identifies elliptic factors.	Partial	Supplied v1, pp. 14, 18, checked.
Morton (2023)	Theorems 1, 3, 5 prove the exact four identities at $N = 5, 7, 2, 3$, respectively, on their full stated domains.	Full for four; partial overall	Supplied v1, pp. 3, 12, 21; final theorem pagination not verified.
Morton (2024)	Tate-form E_7 linear/cubic factor counts; quadratic/sextic assertions are explicitly conjectural in the abstract. This is related to Nakaya's Conjecture 5, not an all-level Conjecture 4 proof.	Partial	Original arXiv v5 abstract checked; full theorem audit not verified.
Jordan (supplied manuscript)	Section 2 and Proposition 5.1 give the coarse quotient and integral coordinate; Corollaries 1.3 and 6.3 give degrees and quadratic-factor counts. Appendix A reconstructs relations and audits multiple plane points.	Partial	Supplied text and code checked; no claim of publication status.
Aricheta (2019)	Theorem 3.1 relates the number of nonrational supersingular points to a genus difference; Remark 1.2 and Appendix A settle the relevant complete-splitting lists. Not the corrected resultant multiplicity formula.	Partial	Original SIGMA paper, pp. 3, 7–8 and relevant sections checked.
Hajouji (2023)	Lemmas 2.6–2.7 and Propositions 2.8–2.9 treat automorphism weights, Fricke orbits and $\mathbb{F}_{p^2}$; Sections 2.4–2.5 use these quadratic relations and resultant radicals and settle Conjecture 2. The fixed-point versus two-point loop distinction is explicit.	Partial	arXiv v3, pp. 6–9 checked. No all-level correction identity found.
Beneish & Wen (2026)	Proposition 3.16 gives the supersingular fixed-point count used above for $D = N, m = 1$; Theorem 6.5 gives quotient special fibres and genera.	Partial	Original arXiv v1, p. 15 and Theorem A statement checked. No resultant formula located.
Sakai (2011)	Atkin orthogonal polynomials for Fricke levels at most 3 and their relation to supersingular polynomials.	Partial background	Publisher abstract/preview checked; full theorem audit not verified.

Source	What was established or inspected	Overlap	Verification
Sakai (2014)	Atkin polynomials for levels 5 and 7, extremal quasimodular forms, and a conjectural supersingular connection.	Partial background	Publisher abstract checked; full theorem audit not verified.
Tsutsumi (2007)	Low-level congruence/triangle-group Atkin polynomials and supersingular elliptic curves.	Partial background	Publisher abstract and metadata checked; full text not verified.
Kaneko & Zagier (1998)	Level-one supersingular polynomials via modular forms, hypergeometric series and Atkin orthogonal polynomials.	Partial background	Author-hosted paper introduction checked; not an all-level Fricke resultant theorem.
Nakaya (2023 preprint; 2024 journal article)	Atkin-like orthogonality and generalized Faber expansions at level one. A visible citing work of Morton, but not evidence for the proposed extension.	None to this identity	arXiv introduction checked; full exclusion/ journal pagination not verified.

Recommendation: a proposition plus a remark. Place the norm-divisor proposition after Jordan's Proposition 5.1 and move the full computational data into an appendix or supplement. The local norm rule is standard, and the branch/elliptic corrections are a natural consequence once the model and coordinate are set up. The explicit uniform packaging at the eleven additional levels appears useful; an unqualified claim that it is the first such result is not supported by this search. No part of Morton's original four-case theorem should be described as still conjectural.

Suggested wording for Jordan's paper:

> The same moduli interpretation also determines the multiplicities in Nakaya's resultant. With the unramified elliptic factors removed, the norm-divisor calculation below gives a uniform identity at every prime level in $\mathfrak{S}$. It recovers the four identities of Nakaya's Conjecture 4, already proved by Morton, and supplies their analogues at the remaining levels. The argument uses standard tame ramification and Fricke fixed-point geometry; we make no priority claim for these geometric ingredients. Exact computations through $p \leq 400$ are included as checks of the formulas and normalizations.

The supplied draft has not been modified. The theorem and proof are supplied as a separate editable LaTeX fragment. The proof here is a mathematical derivation from the stated integral modular-curve facts, not a conclusion drawn from finite agreement; no unresolved plane-normalization or collision lemma is being left implicit. The unresolved issues are bibliographic: final journal theorem pagination, inaccessible full texts, and exhaustive priority.

A All flagged primes in the tested range

All entries below pass. A prime is listed if any certificate described in Section 4 vanishes modulo that prime.

N	Flagged primes $5 \le p \le 400$, $p \neq N$
2	5, 7, 11
3	5, 7, 11, 23
5	7, 11, 19, 59, 71
7	5, 11, 17, 19, 23, 47, 83, 131
11	5, 7, 17, 19, 23, 29, 31, 43, 107, 167, 263, 347, 359
13	5, 7, 11, 17, 23, 43, 47, 107, 251, 311
17	5, 7, 11, 13, 19, 23, 31, 43, 47, 59, 67, 83, 191, 383
19	5, 7, 11, 17, 23, 29, 31, 41, 47, 53, 59, 67, 71, 83
23	5, 7, 11, 13, 17, 19, 31, 41, 43, 47, 53, 67, 79, 83, 107
29	5, 7, 11, 13, 17, 19, 23, 31, 43, 47, 59, 67, 71, 79, 83, 103, 107, 179
31	5, 7, 11, 17, 19, 23, 29, 41, 43, 47, 59, 71, 79, 83, 89, 107, 179, 191, 383
41	5, 7, 11, 17, 19, 23, 29, 31, 37, 43, 47, 59, 67, 71, 79, 83, 103, 107, 127, 131, 139, 151, 163, 239, 383
47	5, 7, 11, 17, 19, 23, 29, 31, 41, 43, 59, 67, 71, 79, 89, 103, 107, 113, 127, 137, 139, 151, 163, 179, 227, 251, 347
59	5, 7, 11, 17, 19, 23, 29, 31, 41, 43, 47, 53, 67, 71, 79, 83, 101, 107, 113, 127, 139, 149, 151, 163, 167, 173, 191, 199, 211, 223, 227, 239, 251, 311, 359
71	5, 7, 11, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 67, 79, 83, 89, 101, 103, 113, 127, 131, 137, 139, 149, 163, 167, 197, 199, 211, 223, 227, 239, 271, 283

B Correction-polynomial output and file map

All fifteen A_N, B_N, E_N are printed in `results/run.log`, in `results/correction_polynomials.txt`, and in the exact GP vectors `results/data_N.gp`. The latter have order

$$[A_N, B_N, E_N, H_N, U_{N,0}, U_{N,1728}].$$

The complete factorizations over $\mathbb{Z}$ of $R_N(0, Y)$, $R_N(1728, Y)$ and D_N appear in `integer_factors_N.txt`. Large coefficients are left in these machine-readable files to avoid transcription errors.

File	Contents
`results/pairs.csv`	One row per permitted pair; status, degrees, maximum multiplicity, timing.
`results/status_matrix.csv`	Requested $N \times p$ PASS table; diagonal marked OUTSIDE.
`results/factors_N.txt`	Full actual and predicted factor matrices for each prime, including the first mismatch field.

File	Contents
`results/radicals_N.gp`	All monic supersingular radical polynomials.
`results/relation_N.gp`	Reconstructed $[a_N, b_N, k, m, v_{\text{sum}}, v_{\text{prod}}]$.
`results/exceptions.csv`	Collision flags for all 246 affected pairs.
`results/exceptional_details.txt`	Polynomial gcds explaining each flag. Units in GP gcd output mean no common root.
`results/certificates.txt`	Exact integer discriminants and pairwise resultants; detects further possible collision primes beyond the survey.
`results/outside_scope.*`	All 43 exploratory pairs, including full factors and the separate formal diagonal diagnostic.
`results/audit.json`	Structural checks of the saved output, pair counts, statuses and exception counts.
`code/verify_c4.gp`	Self-contained reconstruction and multiplicity audit, pinned to GP 2.15.4.
`report/statement_and_proof.tex`	Editable theorem, definitions, and complete geometric proof.

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